

Retrofitting earth-quake damaged housing in Nepal





At least 200,000 damaged houses could be retrofitted.

A further 200,000 homes that have been rebuilt already urgently need retrofitting to make them safe.

At least 30 million tonnes of new building materials could be saved if buildings are retrofitted instead of rebuilt from scratch.

Retrofitting 200,000 houses would save at least \$1.6bn of people's money (\$8,000 per house cheaper than new construction).

Preserving cultural heritage of Nepal

The social and cultural identity of communities could be at risk by large scale demolition / new build



A photograph of a traditional Nepalese house. The house has a steep, thick thatched roof made of dried grass or straw. The walls are made of reddish-brown mud-brick, showing signs of age and wear. There are several windows with dark wooden frames; some are open, and some have shutters. A small wooden structure is attached to the side of the house. The house is situated in a rural area with green trees and vegetation in the background.

Role for traditional skills – like
thatch roofing and carpentry

Highly valued by tourists – who could bring real incomes to beautiful rural communities – traditional artisan skills are being lost as modern construction replaces centuries old vernacular. Retrofitting means celebrating – and preserving - existing cultural heritage of Nepal

- All these houses assessed as category 3 damage.
- To access the housing grant, these families would have to demolish these homes and start again with new foundations.



Dumja VDC, Sindhuli district

Vernacular architecture has adapted to local agricultural economy and life-styles. Many parts of these large houses are used for storage, animals, shops, extended families, etc.



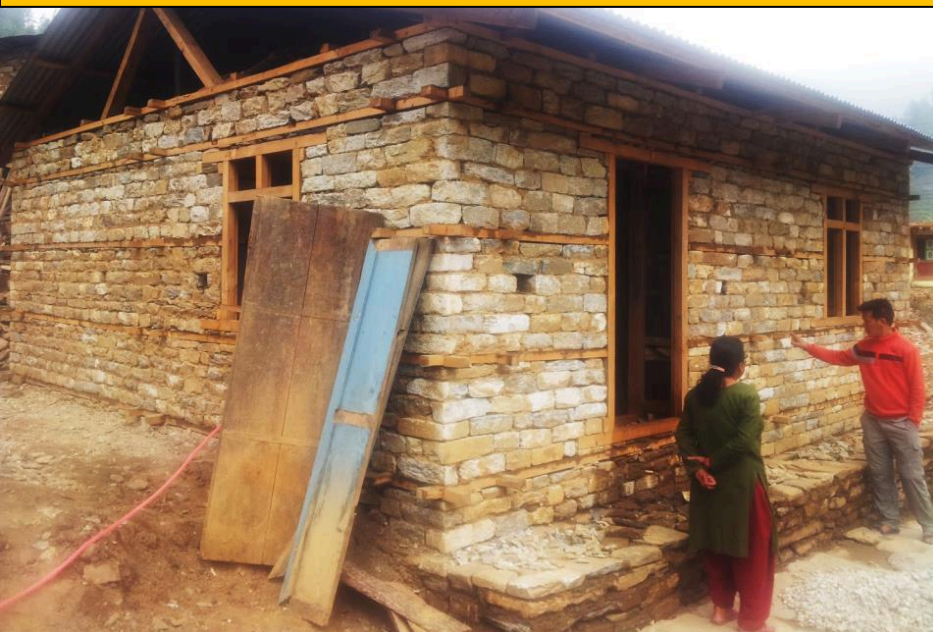
Cost to retrofit:
Approx. \$2,500

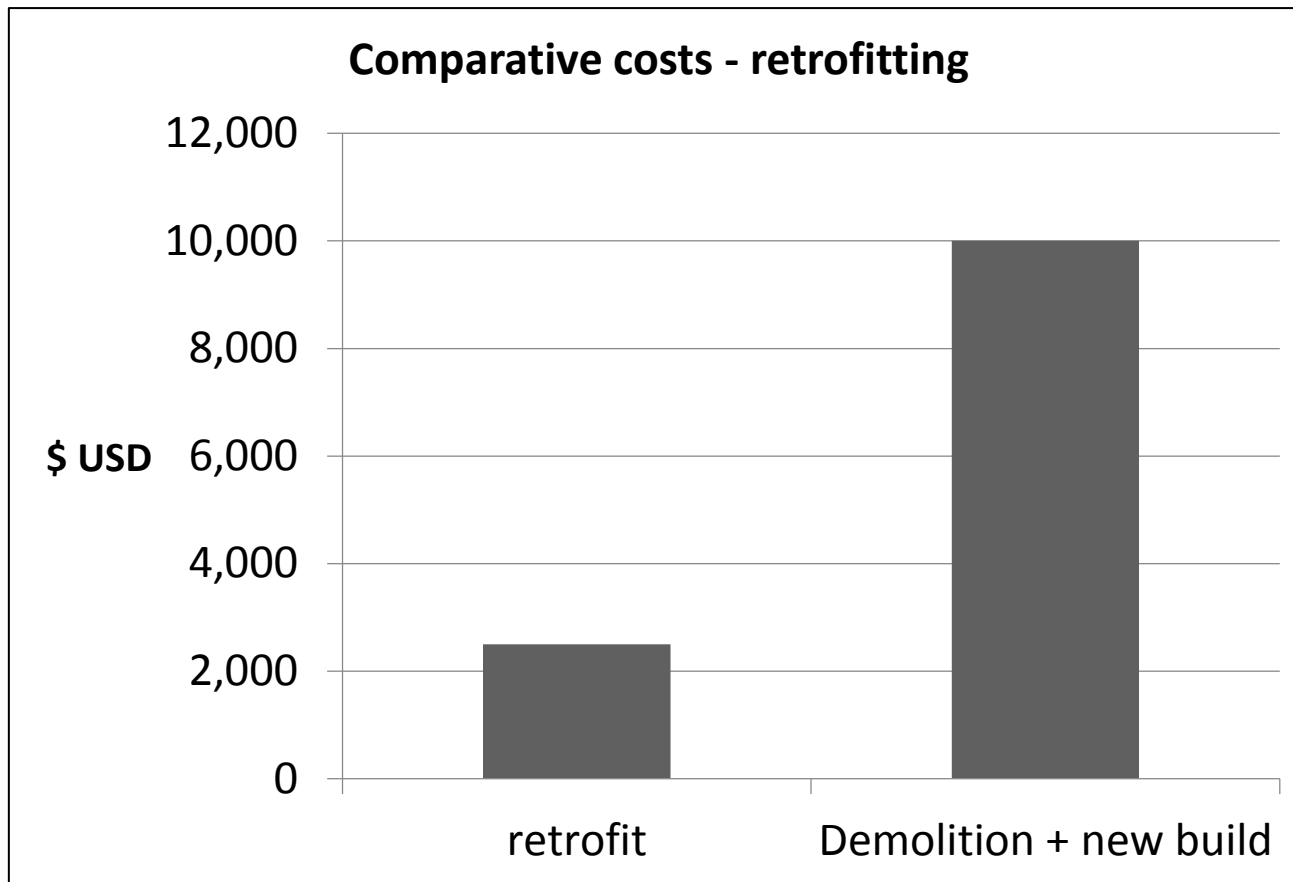
Another example of a category 3 damaged house – which may be suitable for retrofitting.



Compare the size and design of these houses with traditional architecture. What impact for rural Nepal?

Cost for new build like this, following approved designs: < \$10,000





- Up to 250,000 houses may be suitable for retrofitting.
- The saving of around \$8,000 per family is significant.
- Overall this is equivalent to \$2 bn – that could be saved

District	Grade I - II	Grade III	Grade IV	Grade V
Dhading	12,477	15,218	26,080	35,346
Dolakha	4,346	6,520	14,409	35,364
Gorkha	12,906	15,507	22,536	27,125
Kavrepalanchok	20,056	25,130	28,974	23,858
Makwanpur	53,675	18,525	10,928	7,865
Nuwakot	5,360	6,209	23,284	40,295
Okhaldhunga	16,566	9,220	7,946	5,620
Ramechhap	9,278	16,945	20,910	11,490
Rasuwa	620	782	2,064	9,178
Sindhuli	27,557	17,086	15,916	8,191
Sindhupalchok	3,242	3,271	10,796	71,432
Grand Total	166,083	136,413	183,845	275,764

Legend

— International Boundary
— District Boundary

Damage Grade(s)

Grade I - II
Grade III
Grade IV
Grade V
No data available

762,105

private structures

Note that up to 200,000 (~70%) of Grade V damage may have already been rebuilt. And will need retrofitting to be safe.

These charts indicate the damage level by grade in 11 of 14 most affected districts as reported by Nepal Reconstruction Authority (NRA) household survey conducted by Central Bureau of Statistics (CBS).

Map Doc Name: DmgLevel_Graded_Hdwr_000196
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Map Data Source: NRA/CBS
Geo Data Source: DoS, MoFALD, MoHA Geoportal
Web Resource: <http://hrrnepal.org/mapi/>

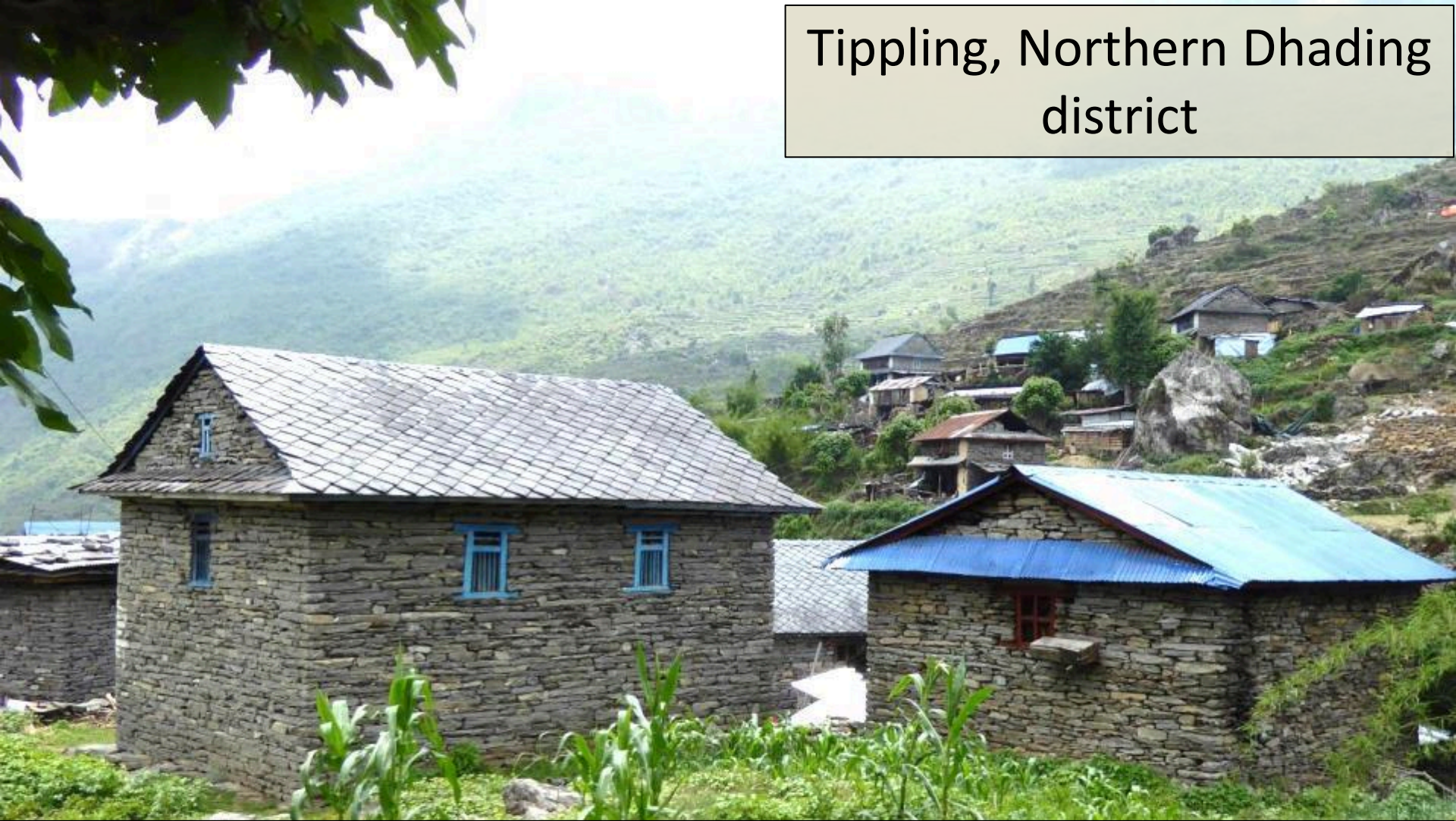


Disclaimer: The boundaries and names shown and the designations used on this map do not imply official endorsement or acceptance by the United Nations



This map shows number of houses per category of damage per district. Where there is less red and more blue and green, it shoes where there is likely more need for retrofitting training, public info and support.

Tippling, Northern Dhading district



- Many families have already rebuilt without reinforcement banding
- < 200,000 estimated to have already rebuilt – will need retrofitting to be safe.

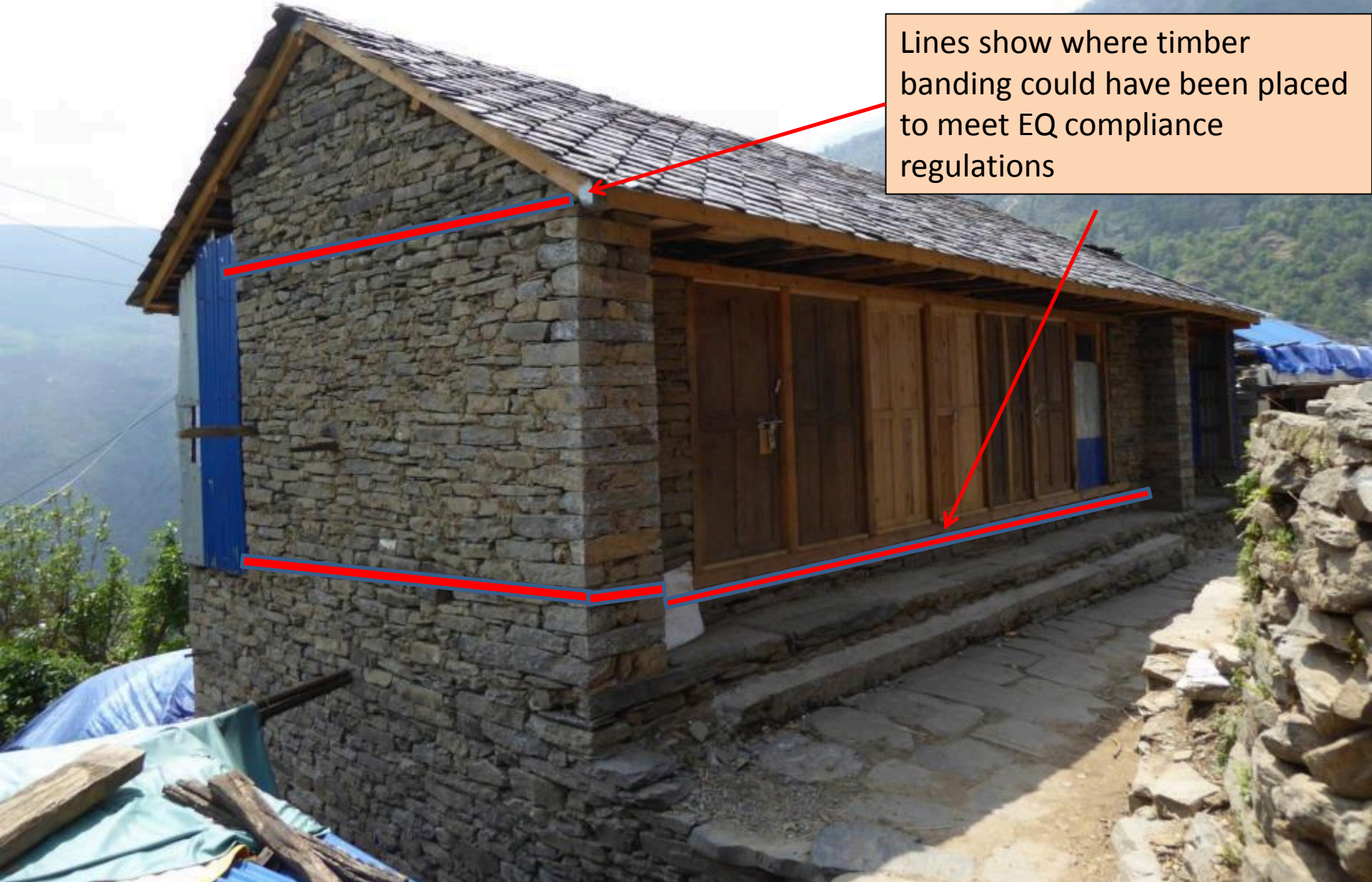
Northern Gorkha (CRS)





Reconstructed home in Haku (Rasuwa). Highly vulnerable to future earthquakes or tremors.

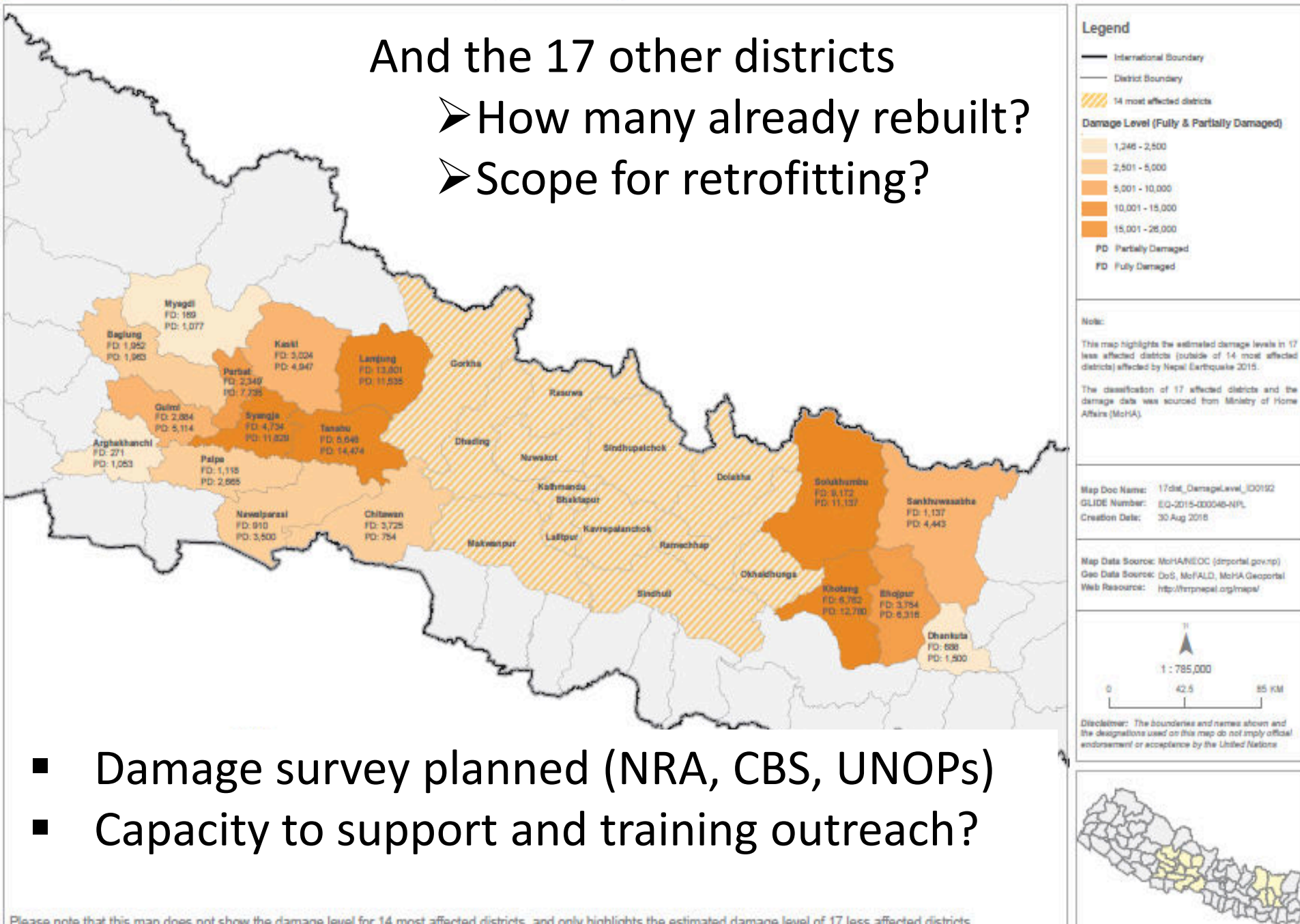
- North Dhading – Lapa VDC. Up to 70% reconstruction already
- EQ compliance (building code) not followed.



Lines show where timber banding could have been placed to meet EQ compliance regulations

And the 17 other districts

- How many already rebuilt?
- Scope for retrofitting?



- Damage survey planned (NRA, CBS, UNOPs)
- Capacity to support and training outreach?